

DFW Housing Market Analysis

Data Dictionary

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Project Type: End-to-End Data Analytics Portfolio Project

Analysis Period: May 2016 – May 2026

Tools Used: Google BigQuery, SQL, Tableau, Microsoft Excel

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1. Purpose

The purpose of this Data Dictionary is to provide a comprehensive reference for every dataset, field, and calculated metric used throughout the DFW Housing Market Analysis project.

This document promotes consistency, transparency, and reproducibility by defining each data element used in the SQL analysis and Tableau dashboards.

2. Project Overview

This project analyzes residential housing market trends across the Dallas–Fort Worth metropolitan area to determine which cities provide the best balance of affordability, historical appreciation, and buyer opportunity.

The project consists of four analytical datasets created from the Zillow Home Value Index (ZHVI). These datasets support three interactive Tableau dashboards and the accompanying executive documentation.

3. Dataset Inventory

Dataset	Purpose
latest_home_values.csv	Current housing market values by city
city_appreciation.csv	Ten-year appreciation by city
county_appreciation.csv	Average appreciation by county
buyer_opportunity_index.csv	Composite ranking of buyer opportunities

4. Dataset 1 – latest_home_values.csv

Purpose

Provides the most recent estimated home value for each city included in the analysis.

Field Name	Data Type	Description	Example
City	Text	Name of the city	Frisco
County	Text	County where the city is located	Collin
State	Text	State abbreviation	TX
Current Home Value	Currency (Decimal)	Latest Zillow Home Value Index estimate	\$645,320

Dataset Usage

Used in:

- Current Market Snapshot Dashboard

- KPI calculations
 - County averages
 - Highest and lowest value city analysis
 - Bubble map visualization
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5. Dataset 2 – city_appreciation.csv

Purpose

Measures the percentage increase in home values for each city between May 2016 and May 2026.

Field Name	Data Type	Description	Example
City	Text	City name	Mansfield
County	Text	County name	Tarrant

Appreciation Percent Percentage Ten-year home value appreciation 128.47%

Dataset Usage

Used in:

- Home Value Appreciation Dashboard
 - Appreciation KPIs
 - Highest appreciating cities
 - Lowest appreciating cities
 - Geographic appreciation map
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6. Dataset 3 – county_appreciation.csv

Purpose

Provides the average appreciation percentage for each county within the study area.

Field Name	Data Type	Description	Example
County	Text	County name	Denton
Average Appreciation Percentage		Mean appreciation across cities in the county	116.82%

Dataset Usage

Used in:

- County comparison chart
- County performance analysis
- Geographic market comparisons

7. Dataset 4 – buyer_opportunity_index.csv

Purpose

Combines affordability and appreciation into a single score that ranks cities according to overall buyer opportunity.

Field Name	Data Type	Description	Example
City	Text	City name	Forney
County	Text	County name	Kaufman
Current Home Value	Currency (Decimal)	Latest estimated home value	\$348,900
Appreciation Percent	Percentage	Ten-year appreciation	142.63%
Appreciation Rank	Integer	Ranking based on appreciation	5
Affordability Rank	Integer	Ranking based on current home value	17
Buyer Opportunity Score	Decimal	Composite score combining affordability and appreciation	22.0

Field Name	Data Type	Description	Example
Buyer Opportunity Rank	Integer	Overall opportunity ranking	3

Dataset Usage

Used in:

- Buyer Opportunity Dashboard
- Opportunity KPIs
- Top Opportunity Cities
- Lowest Opportunity Cities
- Opportunity Bubble Map

8. Calculated Metrics

Average Home Value

Description: Calculates the mean current home value across all analyzed cities.

Business Purpose: Establishes a regional benchmark for comparison.

Highest Home Value

Description: Identifies the city with the highest current home value.

Business Purpose: Highlights premium housing markets.

Lowest Home Value

Description: Identifies the city with the lowest current home value.

Business Purpose: Highlights the most affordable markets.

Cities Above Average

Description: Counts cities whose home values exceed the regional average.

Business Purpose: Measures market distribution.

Average Appreciation

Description: Calculates the average percentage appreciation across all cities.

Business Purpose: Provides a benchmark for historical growth.

Highest Appreciation

Description: Identifies the city with the greatest ten-year appreciation.

Business Purpose: Highlights the strongest-performing housing market.

Lowest Appreciation

Description: Identifies the city with the lowest appreciation.

Business Purpose: Identifies slower-growing markets.

Buyer Opportunity Score

Description: Composite score combining affordability and appreciation rankings.

Business Purpose: Provides a balanced measure of investment potential for prospective buyers.

Buyer Opportunity Rank

Description: Orders cities according to Buyer Opportunity Score.

Business Purpose: Simplifies market comparison for decision-makers.

9. Data Relationships

The project datasets are connected through common geographic identifiers.

Common Fields

- City
- County

These shared fields enable consistent comparisons across datasets and support interactive filtering within Tableau.

Relationship Overview

- **latest_home_values.csv** supplies current market metrics.
- **city_appreciation.csv** provides historical growth data.
- **county_appreciation.csv** summarizes appreciation at the county level.
- **buyer_opportunity_index.csv** integrates affordability and appreciation into a unified ranking.

Together, these datasets support all three Tableau dashboards.

10. Data Quality Notes

Time Period

May 2016 – May 2026

Geographic Scope

Dallas–Fort Worth Metropolitan Area

Data Source

Zillow Home Value Index (ZHVI)

Validation Performed

The following validation steps were completed:

- Verified city and county consistency.
- Confirmed SQL calculations matched Tableau outputs.
- Reviewed KPI values for accuracy.

- Validated dashboard filters and interactions.
 - Checked calculated rankings for logical ordering.
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Assumptions

- Zillow Home Value Index values accurately represent typical residential home values.
 - Historical appreciation is calculated using available monthly home value estimates.
 - City-level analysis provides meaningful comparisons across the DFW metropolitan area.
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Limitations

- Analysis is limited to residential home values.
 - Results are based on historical market performance and should not be interpreted as forecasts.
 - Market conditions may change after the analysis period.
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11. Summary

This Data Dictionary documents the datasets, fields, and calculated metrics used throughout the DFW Housing Market Analysis project.

By providing standardized definitions and technical documentation, this reference improves transparency, supports reproducibility, and ensures consistent interpretation of the data across SQL analysis, Tableau dashboards, and business reporting.

The documented datasets collectively support the project's objective of identifying housing markets that offer the strongest combination of affordability, appreciation, and buyer opportunity within the Dallas–Fort Worth metropolitan area.